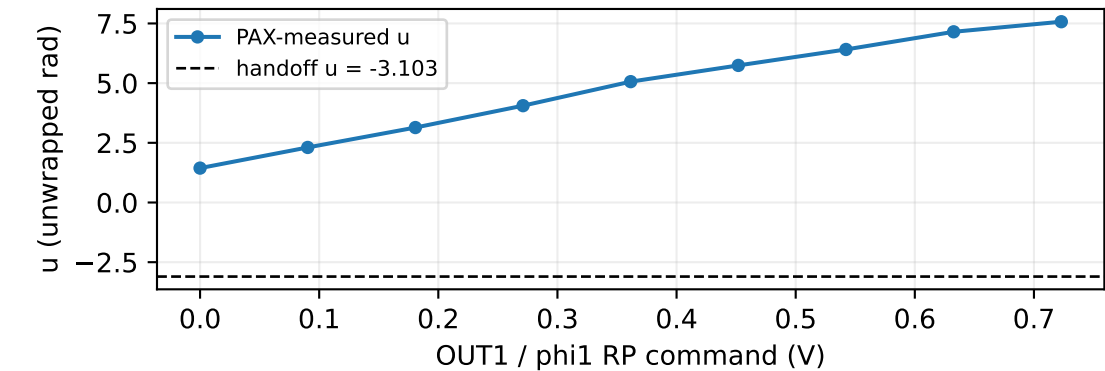
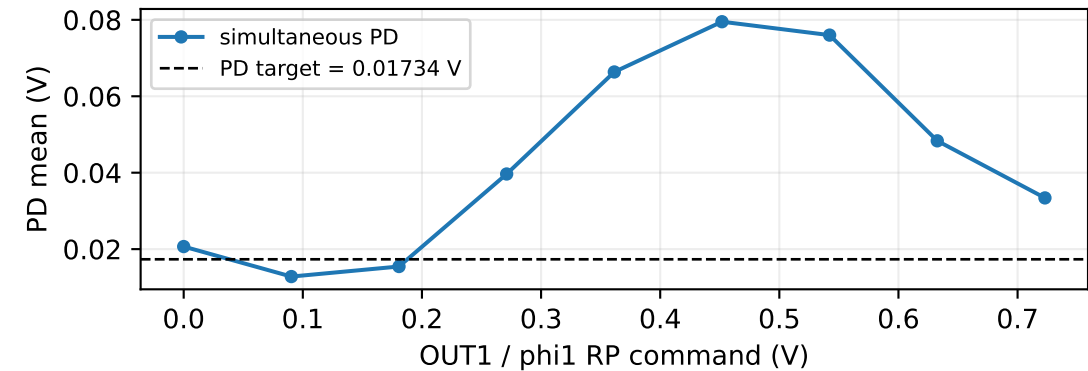


Phi1 PD FPGA lock — 015415_phi1-pd-hybrid-test_phi1-pd-hybrid-2

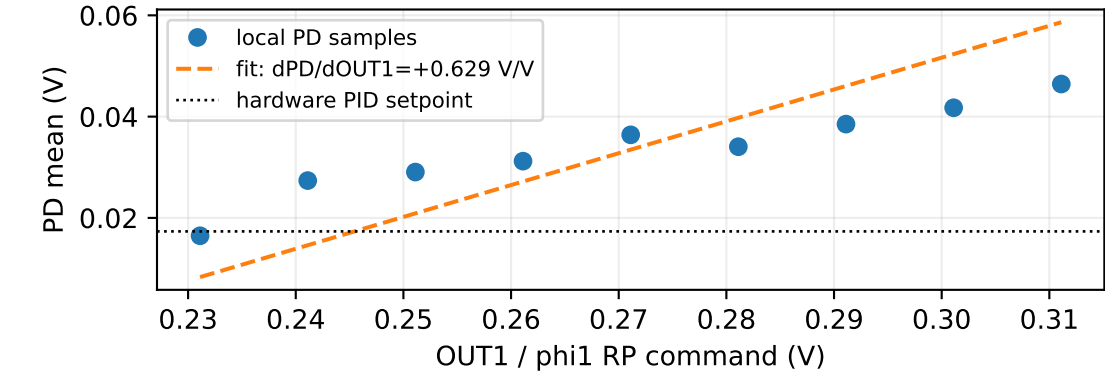
PAX-assisted ϕ_1 rough map



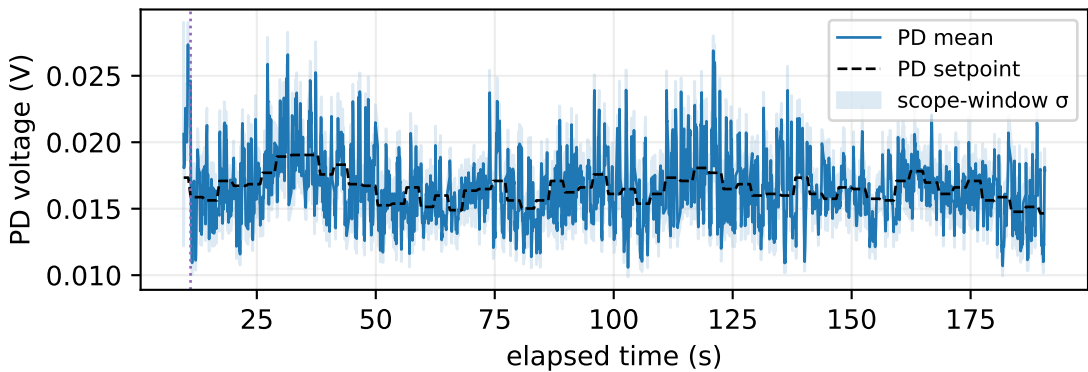
PD value associated with rough u map



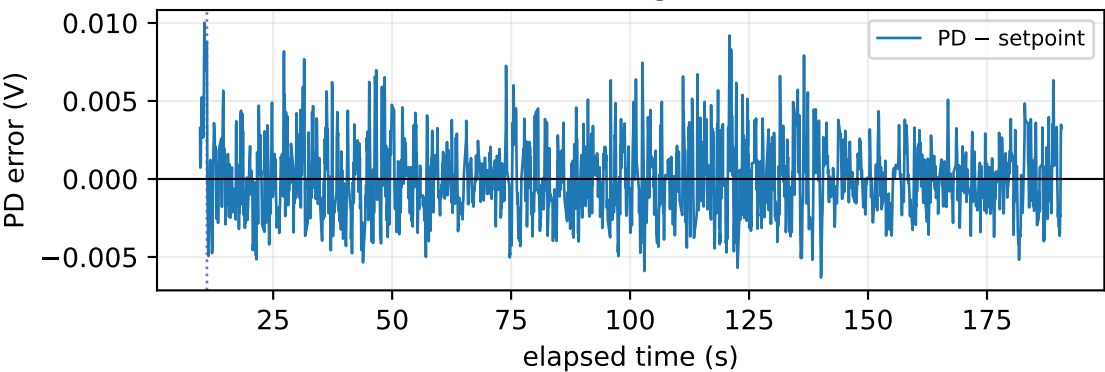
Automatic local PD calibration



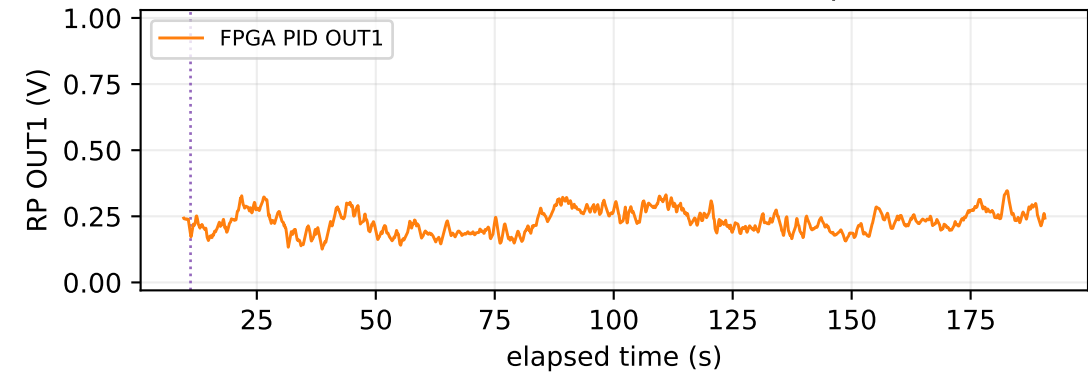
FPGA PD lock



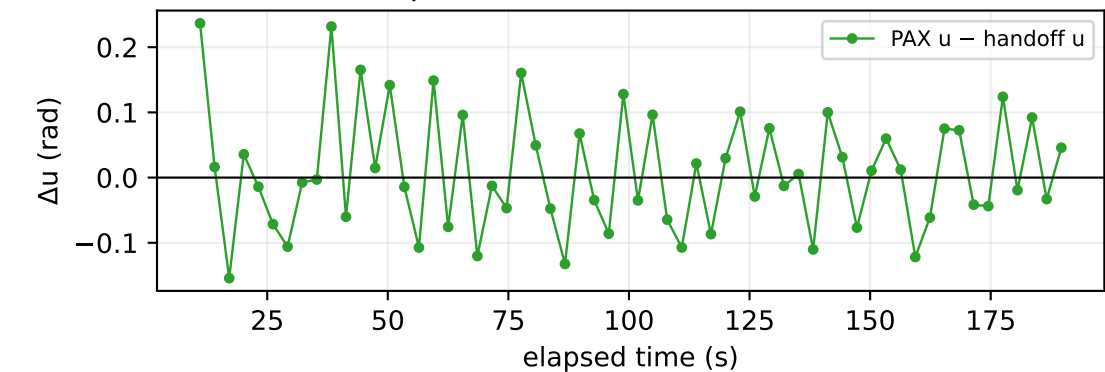
PD locking error



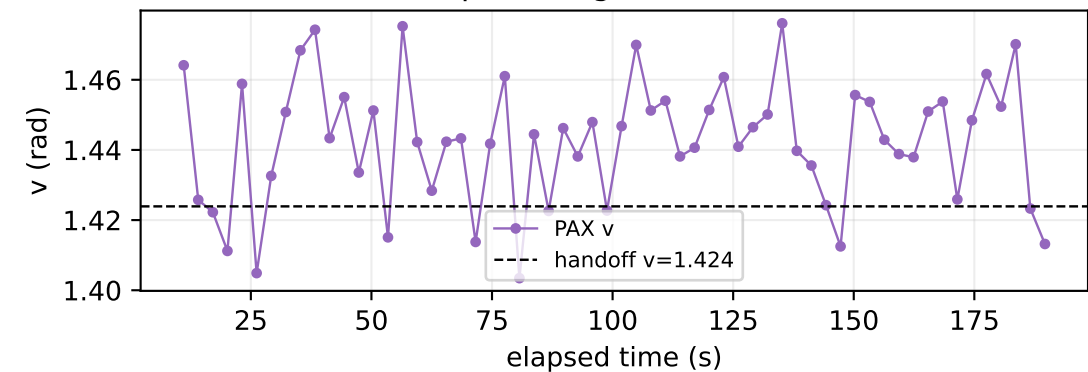
Continuous FPGA actuator output



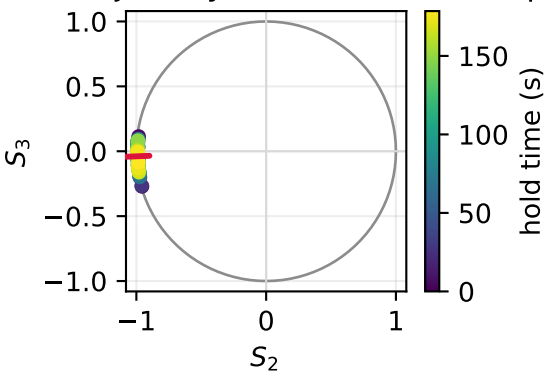
Independent PAX azimuth validation



PAX polar-angle cross-check



Held lock trajectory on the Poincare equator



PAX handoff target $u = -3.1025$ rad | initial PD setpoint = 0.01734 V | local slope = +0.6290 V/V
FPGA PID: P = -0.350, I = -5 Hz | duration = 181.0 s | median |PD error| = 0.00143 V (late 0.00109 V)
OUT1 = 0.126...0.347 V | PD setpoint = 0.01465...0.01904 V | PAX median $|\Delta u|$ = 0.066 rad (60 snapshots)